The University of Nottingham
Malaysia Campus

SCHOOL OF COMPUTER SCIENCE

A LEVEL 1 MODULE, AUTUMN SEMESTER 2016-2017

MATHEMATICS FOR COMPUTER SCIENTISTS

Time allowed ONE Hour

Candidates must NOT start writing their answers until told to do so

There are THREE questions in the exam paper. Answer all the THREE questions.

Only silent, self-contained calculators with a single-line display are permitted in this examination.

Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.

No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.

DO NOT turn examination paper over until instructed to do so

K: knowledge,
C: Comprehension
P: problem solving

1. Questions on Logic, Sets, Functions, and Relations.

(a) Let $A = \{1, 2, 3, 4\}$ and $B = \{2, 4, 5\}$ be two sets and $P(A)$ and $P(B)$ denote their power sets respectively.

(i) Calculate the set difference $A - B$, the set union $A \cup B$, the set intersection $A \cap B$, and the power set $P(B)$, respectively.

[4 marks]

(K) Solutions: $A - B = \{1,3\}, A \cup B = \{1,2,3,4,5\}, A \cap B = \{2,4\}, P(B) = \{\emptyset, \{2\}, \{4\}, \{5\}, \{2,4\}, \{2,5\}, \{4,5\}, \{2,4,5\}\}$

1 mark each.

(ii) Let f be a function from A to B such that $f(1) = 2, f(2) = 4, f(3) = 2, f(4) = 5$. Let g be a function from B to A such that $g(2) = 2, g(4) = 1, g(5) = 4$. For each of these functions determine whether it is one-to-one, onto. Determine the function compositions $f \circ g$ and $g \circ f$ respectively. Show your work.

[6 marks]

(C) Solutions:

f is not one-to-one, but onto. g is one-to-one but not onto.

fog(2)=4, fog(4)=2, fog(5)=5.

Gof(1)=2, gof(2)=1, gof(3)=2, gof(4)=4

1 mark each for the properties and compositions.

(iii) Let $R = \{(a, b) | a \in A \wedge \ b \in A \wedge a \leq (b - 1)\}$ be a subset of the Cartesian product $A \times A$. List all the elements in R.

[6 marks]

(C) Solutions:

R = {(1,2), (1,3), (1,4), (2,3), (2,4), (3,4)}. 1 mark for each element in R.

(iv) Let $R_1 = \{(X, Y) | X \in P(A) \wedge \ Y \in P(B) \wedge X \in Y\}$ and $R_2 = \{(X, Y) | X \in P(A) \wedge \ Y \in P(B) \wedge X \subseteq Y\}$ be two subsets of the Cartesian product $P(A) \times P(B)$. Is R_1 empty? Is R_2 empty? Explain your answers.

[4 marks]

(C) Solutions:
R1 is empty as a would be a subset of A, and b a subset of B, but a can never be an element of b. R2 won't be empty as some of a could be a subset of b, for example, a=null set, and b=B. (1 mark for the correct answer to each of the two questions. 1 mark for the explanation to each of the questions.)

(b) Let A be the set of all people on Facebook. Let R be a relation on the set A such that $\forall a, b \in A$, $(a, b) \in R$ if and only if a shares at least one common friend with b on Facebook.

(i) Is R reflexive, symmetric, and/or transitive? Explain your answers.

[6 marks]

(K, C) Solutions:

R is symmetric since if a has a common friend with b, then b must have the same common friend with a.

R is reflexive since a has the same friends with a. however, am I a friend of myself? If not, if x does not have any friends, (x,x)\not\in R does not. Hence R would not be reflexive.

R is not transitive as a may have a common friend x with b, and b may have a common friend y with c, but x is not necessarily the same as y, thus we cannot say that a definitely has a common friend with c.

(1 mark for the correct answer to each of the two questions. 1 mark for the explanation to each of the questions.)

(ii) When is an ordered pair (a, b) in R^2 ? Explain your answer.

[2 marks]

(C) Solutions:

For (a,b) in R2, there must exist some c such that (a,c) is in R (1 mark) and (c,b) is in R (1 mark).

(iii) Let S be a relation on the set A such that $\forall a, b \in A$, $(a, b) \in S$ if and only if a knows about b. Determine $R \cup S$, $R \cap S$, and $R - S$.

[6 marks]

(K) Solutions:

RUS consists of (a,b) such that either a shares common friends with a, or a knows about b.

R intersection S consists of (a, b) such that a not only shares common friends with b, but also knows about b.

R-S consists of (a,b) such that a shares common friends with b but a does not know about b.

(2 marks each)

2. Questions on Logic, Number Theory, Induction, and Matrices.

(a) If a and b are integers with $a \neq 0$, we say that a divides b if there is an integer c such that $b = ac$. The notation $a \mid b$ denotes that a divides b. If a and b are integers and m is a positive integer, then a is congruent to b modulo m if m divides $a - b$. We use the notation $a \equiv b (\mathrm{mod}\, m)$ to indicate that a is congruent to b modulo m.

(i) Using $\forall, \exists, \neg, \wedge, \vee, \rightarrow$, and/or $\leftrightarrow$, express the above definition "a divides b" in predicate logic.

[4 marks]

(P) Solutions:

$\forall a, b \in Z, a \neq 0, a|b \leftrightarrow \exists c \in Z, b = ac$. 1 marks for using the universal quantifier correctly, 1 marks for using iff correctly, 1 mark for using the existential quantifier correctly, and 1 mark for using b=ac. Any other alternative representation is also acceptable.

(ii) Show that if a, b, c and m are integers such that $m \geq 2$, $c > 0$, and $a \equiv b (\mathrm{mod}\, m)$, then $ac \equiv bc (\mathrm{mod}\, mc)$.

[5 marks]

(C,P) Solutions:

According to the given condition that a is congruent to b modulo m, we know that $m|(a-b)$ (1 marks), which means that there exists some integer n such that $(a-b) = mn$ (1 mark). Multiplying both sides by an integer c>0, we have (a-b)c=mnc (1 mark), hence (ac-bc)=(mc)n (1 mark), so it shows that ac is congruent to bc modulo mc (1 mark).

(iii) Let $a = bq + r$, where a, b, q and r are integers. The Euclidean algorithm says that the greatest common divisor of a and b is the same as that of b and r, denoted as $\gcd(a, b) = \gcd(b, r)$. Find the greatest common divisor of 91 and 287 using the Euclidean algorithm.

[5 marks]

(K) Solutions:

Gcd(287,91) = gcd(91,14) = gcd(14,7) = gcd(7,0) = 7 (1 mark for each step)

(b) A sequence is an ordered list. For example, $3, 5, 8, 12, 17, 23, 30, 38, 47, \ldots$ is an infinite sequence.

(i) Provide a simple formula that generates an integer sequence that begins with the above given list.

[4 marks]

(K,P) Solutions:

a(1) = 3, a(n) = a(n-1)+n for all n>=2. (1 mark for the initial term, 3 marks for the general formula, any intermediate steps leading the final formula will be given marks.)

(ii) Let $\{a_n\}$ denote the above given sequence, where $n \geq 1$ are integers and $a_1 = 3$. Let $S_m = \sum_{j=m}^{2m-1} a_j$, where $m \geq 1$ are integers. List the first **THREE** terms S_1, S_2, and S_3 of the sequence $\{S_m\}$. Show your work.

[4 marks]

(C,P) Solutions:

S1 = a1 = 3 (1 mark), S2 = a2+a3=5+8=13 (1 mark), S3 = a3+a4+a5 = 8+12+17=37 (2 marks). Alternatively marks will be awarded if any intermediate steps are given leading toward the final answer.

(iii) Suppose that $\boldsymbol{A}$ and $\boldsymbol{B}$ are square matrices with the property $\boldsymbol{AB} = \boldsymbol{BA}$. Using the mathematical induction show that $\boldsymbol{AB}^n = \boldsymbol{B}^n\boldsymbol{A}$ for every positive integer n.

[6 marks]

(K,C) Solutions:

Proof for the basic step: for n=1, AB=BA by given conditions. (1 mark)

Proof for the inductive step: assume when n=k>1, AB^n = B^n A (1 mark). We have AB^(n+1) = A(B^n B) = (AB^n)B = (B^n A) B = B^n (BA) = (B^nB)A = B^(n+1)A. Shown. (0.5 marks for each intermediate step)

(iv) Let $P(n)$ be the statement that $[(p_1 \to p_2) \wedge (p_2 \to p_3) \wedge \ldots \wedge (p_{n-1} \to p_n)] \to [(p_1 \wedge p_2 \wedge \ldots \wedge p_{n-1}) \to p_n]$ is a tautology, where n is an integer. We can prove by mathematical induction that $P(n)$ holds whenever $p_1, p_2, \ldots, p_n$ are propositions, where $n \geq 2$. Specify the statement that is used in the basic step of the mathematical induction and prove that it is true.

[5 marks]

(K,C) Solutions:

The basic step is to prove that when n=2 that P(2) holds (1 mark). This is to show that (p1->p2)->(p1->p2) is true (1 mark). This is the same as to prove that q->q is true (1 mark). We know that q->q is the same as (not q) or q (1 mark), and this is a trivial tautology (1 mark).

3. Questions on Sets, Functions, Counting, and Probability.

(a) Let $A = \{1, 2, 3\}$ and $B = \{a, b, c, d, e\}$ be two sets, and a function $f: A \rightarrow B$ is defined such that every element in A is mapped to an element in B.

(i) How many such functions are there from A to B? Show your work.

[4 marks]

(K,C) Solutions:
There are three elements in A, each of them can be assigned one of the 5 elements in B. thus the total different number of functions is 5*5*5=125. 2 marks for recognising the choice of different assignment is a sum rule, 2 marks for recognising that the production rule is used to form a function from all elements in the domain.

(ii) Among the results in (i), how many functions are one-to-one? Show your work.

[5 marks]

(C,P) Solutions:
In order for f to be one-to-one, they have to be assigned with 3 different elements in B. There are C(5,3) ways of choosing 3 out of 5 elements. However, the different orders of the assignments of the elements in B to the elements in A will result in different functions, thus it should be C(5,3)*P(3,3) = P(5,3). (2 marks for recognising C(5,3), 2 marks for recognising P(3,3), 1 mark for the final formula.)

(iii) Among the results in (i), how many functions are there that assign a to exactly one of the positive integers less than 3? Show your work.

[6 marks]

(C,P) solutions:

a will be assigned to either 1 or 2 in A (2 marks). We then any alphabet out of the rest of the 4 elements in B to assign to any of the rest of 2 elements in A (2 marks). So the result is 4*4*2 (2 marks).

(iv) Assume that A is given such that $A = \{1, 2, 3, 4, 5, 6, 7, 8\}$. B and f remain the same as specified above. Using the Pigeonhole Principle show that f is not a one-to-one function.

[6 marks]

(C) Solutions:
Now A has 8 elements (pigeons, 2 marks) but B still has only 5 elements (pigeonholes, 2 marks). So according to pigeonhole principle, there will be at least one element in B that is assigned to 2 elements in A (since ceiling(8/5)=2, 2 marks). Thus f is not one-to-one.

(b) What is the probability of the following events when we randomly select a permutation of the 5 lowercase letters of the English alphabet $\{a, b, c, d, e\}$?

(i) The first 3 letters of the permutation are in alphabetical order.

[6 marks]

(C,P) Solutions:

There are in total P(5,5) permutations (1 mark). To keep the 3 letters in alphabetic order, it is to choose 3 out of 5 (or equivalently 2 out of 5) C(5,3) (2 marks), and the rest two letters take a permutation 2! (2 marks). The final result is then C(5,3)*2!/5! (1 mark).

Other alternative solutions will be awarded marks accordingly.

(ii) a and e are next to each other in the permutation.

[6 marks]

(C,P) Solutions:
Since a and e are bound together, there are effectively 4 alphabets for in permutation P(4,4) (3 marks). However, since changing the order of a and e will result in a different order (2 marks), so we have P(4,4)*P(2,2)/P(5,5) (1 mark).